manufacture of explosives. In order to harness this energy for peaceful purposes, it is essential to be able to produce a chain reaction which can be kept under control, so as to permit the withdrawal of just the right quantity of energy required at any given moment. Fortunately, it is possible to produce such a chain reaction in natural uranium, which is a mixture of two uranium isotopes—$_{92}U^{238}$ and $_{92}U^{235}$, in the proportion of 140 to 1, and it is not necessary to enrich the natural uranium in the rare $_{92}U^{235}$ first.

However, a chain reaction does not take place in pure metallic uranium, for the neutrons emitted in the fission reaction collide far more frequently with $_{92}U^{238}$ nuclei than with $_{92}U^{235}$ nuclei. They are reflected by the $_{92}U^{238}$ nuclei, as a rule, without loss in velocity, and are eventually captured by one of these nuclei with which they resonate, so that they are lost for the chain reaction. But pieces of uranium can be encased in a *moderator,* a substance that slows down the neutrons. This permits the neutrons to be carried quickly past the resonance values and their speed to be reduced to thermal velocities. But despite the relatively small number of $_{92}U^{235}$ nuclei, the thermal neutrons are more easily captured by them than by the $_{92}U^{238}$ nuclei and, as a rule, they then produce fission in the $_{92}U^{235}$ nuclei. The effect of the moderator thus is that the neutrons are seldom captured by $_{92}U^{238}$ nuclei. If the moderator selected is a substance which absorbs thermal neutrons to a very small extent only, and if the apparatus is of a sufficient size to prevent the escape of too many neutrons through the surface before entering into reaction with the $_{92}U^{235}$ nuclei, a chain reaction can be started. Practically, the only substances suitable for use as moderators are heavy water (D_20) and absolutely pure graphite, the absorption coefficient of both of which is very small for thermal neutrons. In a uranium *reactor* (as an apparatus consisting of uranium and a moderator is called) built of pieces of uranium and heavy water, the following chain reaction takes place: A neutron, released by fission, leaves the piece of uranium—possibly after a few collisions with uranium atoms—and reaches the heavy water. There, due to collisions with deuterons, it loses velocity, and wanders around in the moderator at thermal velocity, until eventually it happens to collide again with a piece of uranium. It then produces a new fission in a $_{92}U^{235}$ nucleus, as a result of which again two or three neutrons are liberated—and so it goes on. While the chain reaction in the atomic bomb is provoked by fast neutrons, the velocity of which is reduced by inelastic collisions to only a little below their original speed, the chain reaction in the uranium reactor is propagated by slow neutrons.

This chain reaction can be easily controlled and guided. The uranium is heated by the disintegration of the uranium atoms. The result is a spreading of